

VisionGuard **ULTRA**

Smart AI for normal CCTV cameras.

Detect → Decide → Dispatch.

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PROBLEM

Cameras record. Nobody watches.

20M+

people in Karachi

~20 min

until an operator gets tired

Late

help arrives after the crime

SOLUTION

Make every camera smart

Detect

finds danger in real time

Decide

gives a risk score 0–100

Dispatch

alerts 15 · 16 · 115 · 1122

WHAT IT DETECTS

12 types of emergency

 Fire  Fight  Weapon  Robbery  Kidnapping 

Crowd crush

 Car accident  Bike accident  Loitering  Left bag 

Blocked road  Gunshot / scream

HOW IT WORKS

Camera → AI → Rules → Alert

1. YOLOv8s	finds people, cars, bags
2. ByteTrack	gives each person an ID
3. Special models	pose · fire · weapon · violence
4. Event engine	12 rules, checked over time
5. Action	risk score · alert · sealed evidence

MODELS

3 models trained by our team

Fire & Smoke

YOLOv8s · 80 epochs

Weapon

YOLOv8s · 60 epochs

Violence

YOLOv8s-cls · 50 epochs

Pre-trained

YOLOv8s · YOLOv8s-pose · YAMNet audio

Trained on Google Colab T4 GPU using transfer learning.

DATA

~35,000 training images

16,027

fire images (D-Fire)

7,116

weapon images

~12,000

violence images (RWF-
2000+)

4,304

extra weather images

All images scanned · 5 duplicates and 74 bad labels removed.

2× better than our old method

47.6%

precision

old: ~25%

25.5%

recall

16.0%

mAP@50

old: ~8%

7.6%

mAP@50-95

Measured on the D-Fire test set — a hard dataset with many tiny fires. Weapon & violence results: in progress.

SPEED

43% faster with our async design

7 → 10

frames per second

-48%

dropped frames

4 GB

small GPU (Quadro T1000)

BETTER RESULTS FROM REAL DATA

3 settings tuned

Crowd danger level	0.90 → 0.95
Time before alert	1.0 s → 1.5 s
Detection confidence	0.30 → 0.35

Fewer duplicate alerts and fewer false fights in dark video.

GOALS

Goal vs. today

12 event types	● Done
Sealed evidence (SHA-256)	● Done
No face recognition	● Done
Speed 25–30 FPS	● In progress (10 FPS)
Detection mAP@50 $\geq$ 70%	● Not yet (16%)
False alarms < 5%	● Measuring

NEXT STEPS

What we do next

More training

bigger model, more epochs

Local data

real Karachi CCTV

Finish tests

weapon & violence reports

City pilot

real cameras, real helplines

Cameras are already there.
We give them a brain.

Thank you — questions?